2	**(a)**	For an object to move in circular motion at constant speed, it only changes direction hence, the change in velocity must be perpendicular to the direction of its velocity.	**B1**
		Thus, the acceleration needs to be perpendicular to its velocity which leads to the resultant force being perpendicular to the velocity	**B1**
		Alternative:	
		Since the object is moving at constant speed, its KE does not change.	**B1**
		By Work-Energy Theorem, the work done by the resultant force must then be zero. Thus the resultant force must act perpendicular to its velocity.	**B1**

	(b)(i)	$v = r\omega$ $= (6.2)(2\pi/4.1)$ $= 9.5\ \text{m s}^{-1}$	 **C1** **A1**
	(b)(ii)	Acceleration = $r\omega^2$ $= (6.2)(2\pi/4.1)^2$ $= 14.6\ \text{m s}^{-2}$	 **C1** **A1**
	(b)(iii)	R_B – Weight = resultant force $R_B - (75)(9.81) = (75)(14.6)$ $R_B = 1830$ N	 **C1** **A1**
	(c)	$R_A = ma - mg$, $R_A > 0$ is required to remain contact so $v^2/r > g$, i.e. $v > \sqrt{rg}$, the minimum speed for a contact force to be required	**B1** **B1**
	(d)	minimum speed is when $v^2/r = g$ $v^2/6.2 = 9.81$ $v = 7.8\ \text{m s}^{-1}$	 **C1** **A1**

3	**(a)**	There is no lost volts or energy in the internal resistance of the battery *QR*	**B1**